

Lab09 – blue/green deployment

1. Create a Deployment named `nginx-blue` with 3 replicas. The Pod template of the Deployment should use container image `nginx:1.23.0` and assign the label `version=blue`.

```
vagrant@kubernetes-control-plane:~/Lab$ vim blue-deployment.yaml
vagrant@kubernetes-control-plane:~/Lab$
vagrant@kubernetes-control-plane:~/Lab$ cat blue-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-blue
spec:
  replicas: 3
  selector:
    matchLabels:
      version: blue
  template:
    metadata:
      labels:
        version: blue
    spec:
      containers:
      - image: nginx:1.23.0
        name: nginx
        ports:
        - containerPort: 80
vagrant@kubernetes-control-plane:~/Lab$
vagrant@kubernetes-control-plane:~/Lab$ kubectl apply -f blue-deployment.yaml
deployment.apps/nginx-blue created
vagrant@kubernetes-control-plane:~/Lab$

vagrant@kubernetes-control-plane:~/Lab$ kubectl get pods -l version=blue
NAME                                READY   STATUS    RESTARTS   AGE
nginx-blue-f87f459f9-bls6w          1/1     Running   0           2m23s
nginx-blue-f87f459f9-hrkbn          1/1     Running   0           2m23s
nginx-blue-f87f459f9-km29c          1/1     Running   0           2m23s
vagrant@kubernetes-control-plane:~/Lab$
```

2. Expose the Deployment with a Service of type `ClusterIP` named `nginx`. Map the incoming and outgoing port to 80. Select the Pod with label `version=blue`.

```
vagrant@kubernetes-control-plane:~/Lab$ vim service.yaml
vagrant@kubernetes-control-plane:~/Lab$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx
spec:
  selector:
    version: blue
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
vagrant@kubernetes-control-plane:~/Lab$
vagrant@kubernetes-control-plane:~/Lab$ kubectl apply -f service.yaml
service/nginx created
vagrant@kubernetes-control-plane:~/Lab$
```

- Run a temporary Pod with the container image `alpine/curl:8.5.0` to make a call against the Service using `curl`. The returned headers will include the nginx server version.

```
vagrant@kube-control-plane:~/Lab$ kubectl run tmp --image=alpine/curl:8.5.0 --restart=Never
-it --rm -- curl -sI nginx.default.svc.cluster.local | grep Server
Server: nginx/1.23.0
vagrant@kube-control-plane:~/Lab$
```

- Create a second Deployment named `nginx-green` with 3 replicas. The Pod template of the Deployment should use container image `nginx:1.23.4` and assign the label `version=green`.

```
vagrant@kube-control-plane:~/Lab$ vim green-deployment.yaml
vagrant@kube-control-plane:~/Lab$ cat green-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-green
spec:
  replicas: 3
  selector:
    matchLabels:
      version: green
  template:
    metadata:
      labels:
        version: green
    spec:
      containers:
        - image: nginx:1.23.4
          name: nginx
          ports:
            - containerPort: 80
vagrant@kube-control-plane:~/Lab$
vagrant@kube-control-plane:~/Lab$ kubectl apply -f green-deployment.yaml
deployment.apps/nginx-green created
vagrant@kube-control-plane:~/Lab$
vagrant@kube-control-plane:~/Lab$ kubectl get pods -l version=green
NAME                                READY   STATUS    RESTARTS   AGE
nginx-green-6fc95f569d-4hwd5        1/1     Running   0           29s
nginx-green-6fc95f569d-h7m7k        1/1     Running   0           29s
nginx-green-6fc95f569d-nwc59        1/1     Running   0           29s
vagrant@kube-control-plane:~/Lab$ _
```

- Change the Service's label selection so that traffic will be routed to the Pods controlled by the Deployment `nginx-green`.

```

vagrant@kubernetes-control-plane:~/Lab$ vim service.yaml
vagrant@kubernetes-control-plane:~/Lab$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx
spec:
  selector:
    version: green
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
vagrant@kubernetes-control-plane:~/Lab$
vagrant@kubernetes-control-plane:~/Lab$ kubectl apply -f service.yaml
service/nginx configured
vagrant@kubernetes-control-plane:~/Lab$
vagrant@kubernetes-control-plane:~/Lab$ kubectl run tmp --image=alpine/curl:8.5.0 --restart=Never -it --rm -- cu
rl -sI nginx.default.svc.cluster.local | grep Server
Server: nginx/1.23.4
vagrant@kubernetes-control-plane:~/Lab$

```

6. Delete the Deployment named `nginx-blue`. Run a temporary Pod with the container image `alpine/curl:8.5.0` to make a call against the Service.

```

vagrant@kubernetes-control-plane:~/Lab$ kubectl delete deployment nginx-blue
deployment.apps "nginx-blue" deleted
vagrant@kubernetes-control-plane:~/Lab$
vagrant@kubernetes-control-plane:~/Lab$ kubectl get pods -l version=blue
No resources found in default namespace.
vagrant@kubernetes-control-plane:~/Lab$
vagrant@kubernetes-control-plane:~/Lab$ kubectl get pods -l version=green
NAME                                READY   STATUS    RESTARTS   AGE
nginx-green-6fc95f569d-4hwd5        1/1     Running   0           5m39s
nginx-green-6fc95f569d-h7m7k        1/1     Running   0           5m39s
nginx-green-6fc95f569d-nwc59        1/1     Running   0           5m39s
vagrant@kubernetes-control-plane:~/Lab$

```